SUMULONG MEMORIAL HIGH SCHOOL
Gen. Luna St. Brgy. San Roque, Antipolo City
SY. 2022-2023

CIRCULATORY SYSTEM

LEARNING MODULE

NAME: ________________________________ **YEAR AND SECTION :**__________________

SET THE TARGET

LEARNING OBJECTIVES:

At the end of this module, you are expected to:

1. Identify and describe the structure of the organs and tissues composing the cardiovascular system;
2. Explain the mechanism on how the respiratory and circulatory systems work together to transport nutrients, gases, and molecules to and from the different parts of the body.

In this module you will answer the following questions:

- How do the respiratory and circulatory systems work with each other?
- How do the diseases in the circulatory and respiratory systems begin to develop?
- How can a person's lifestyle affect the performance of the respiratory and circulatory systems?

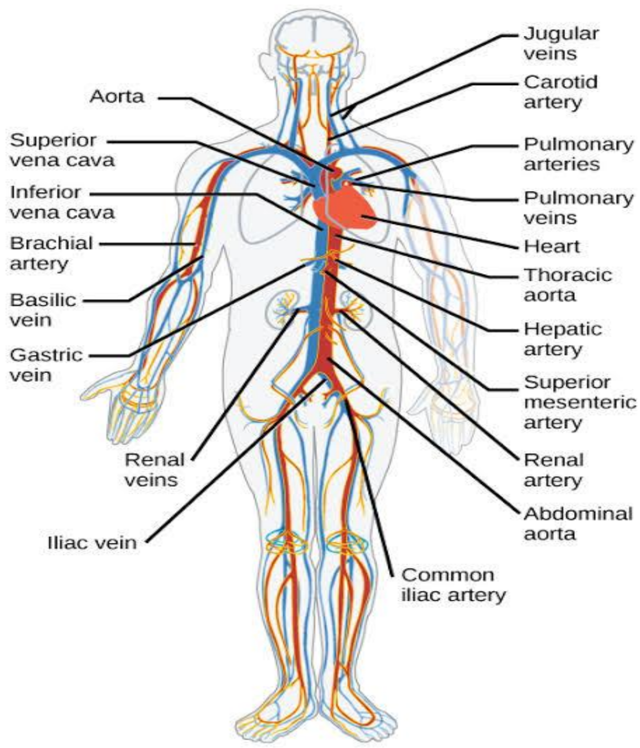
PRE-ASSESSMENT

Direction: Fill in the K-W-H-L Chart below to assess your prior knowledge and understanding of the topic, Respiratory and Circulatory Systems, Working with the other Organ Systems.

K	W	H	L
What do I know?	What do I want to find out?	How can I find out what I want to learn?	What did I learn

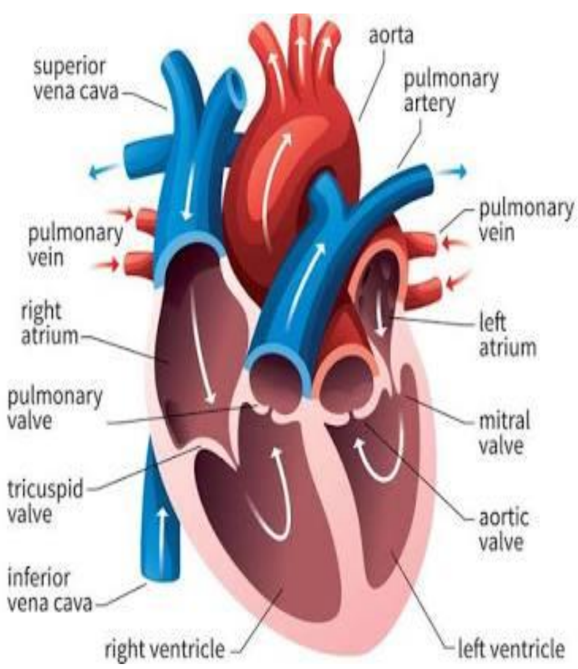
LEARN MORE!

The **circulatory system** is the life support structure that nourishes your cells with nutrients from the food you eat and oxygen from the air you breathe. It can be compared to a complex arrangement of highways, avenues and lanes connecting all the cells together into a neighborhood. Sequentially, the community of cells sustains the body to stay alive. Another name for the circulatory system is the cardiovascular system. The circulatory system functions with other body systems to deliver different materials in the body. It circulates vital elements such as oxygen and nutrients. At the same time, it also transports wastes



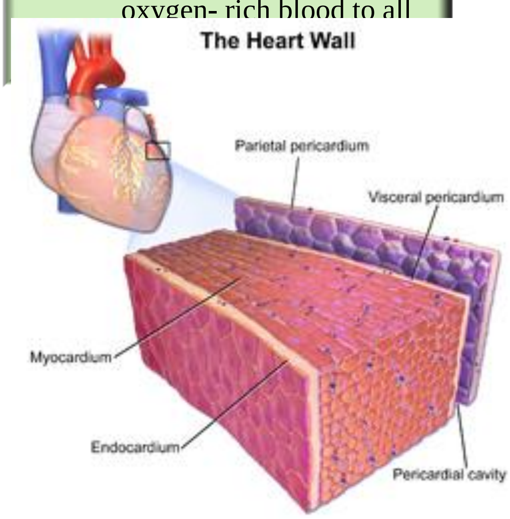
The following are the three major parts of the circulatory system, with their roles:

1. HEART – pumps the blood throughout the body and it consists of four chambers.



Four Chambers of the Heart

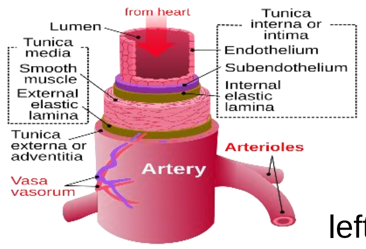
- Right Atrium- receives oxygen- poor blood from the vena- cava
- Left Atrium- receives oxygenated blood from the pulmonary vein.
- Right Ventricle- pumps blood for oxygenation.
- Left Ventricle- pumps oxygen- rich blood to all



THREE LAYERS OF THE HEART

- Epicardium- outer layer of the heart wall.
- Myocardium- middle layer of the heart.
- Endocardium- inner layer of the heart.

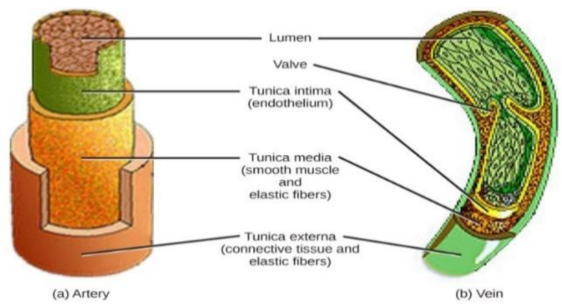
2. Blood vessel – These vessels are connected to the heart that carries the blood throughout the body.



- **ARTERIES -** carry oxygenated blood away from the heart to the cells, tissues and organs of the body. Except for the pulmonary arteries that transport oxygenated blood from the ventricle to the body tissues.
- **AORTA-** largest artery that conveys oxygen- rich blood pumped by the left ventricle to all parts of the body.

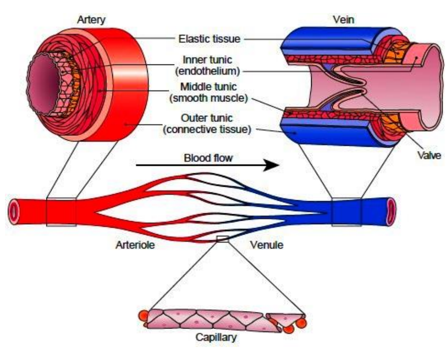
- **ARTERIOLES-** The arterioles are smallest arteries. It play a key role in regulating blood flow into the tissue capillaries.

- **VEINS -** carry deoxygenated blood to the heart. From the capillaries of the different tissues and organs in the body, blood enters the smallest veins called ***venules.***



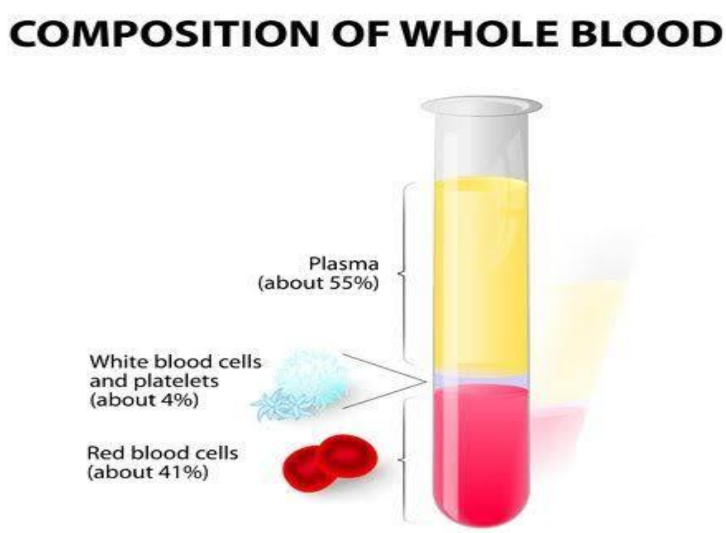
THREE LAYERS OF THE VEINS

- Tunica Adventitia
- Tunica Media
- Tunica Intima



- **CAPILLARIES -** the smallest blood vessels in the body, connecting the smallest arteries to the smallest veins - the actual site where gases and nutrients are exchanged

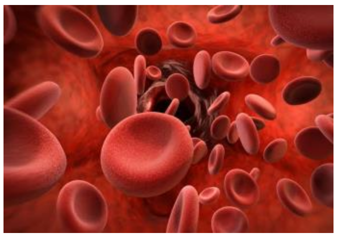
3. BLOOD – The blood is a liquid tissue and the circulating medium of the cardiovascular system. The blood comprises 8% of the human body weight. An average adult has about five liters of blood. Its main functions are: to *transport materials* and *to fight infections.* Blood has two components: the **plasma** and the **corpuscles** that are suspended in the plasma.



- **PLASMA-** The plasma is the liquid portion of the blood. It comprises about 55% of the blood weight. It is a straw colored fluid consisting of about 92% water, 8% blood proteins and the trace amounts of inorganic materials.

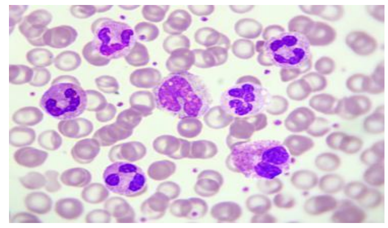
- **CORPUSCLES-** or the formed elements suspended in the plasma, comprise 45% of the blood body weight.

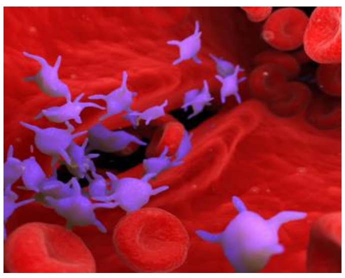
FORMS OF CORPUSCLES



- **ERYTHROCYTES-** are the red blood cells. The red color is attributed to the pigment *hemoglobin,* an iron- containing molecule that can bind with oxygen. This is the reason why red blood cells can transport gases throughout the body. Red blood cells are *biconcave* in shape, elastic and anucleated. Their life span is about 80 to 120 days.

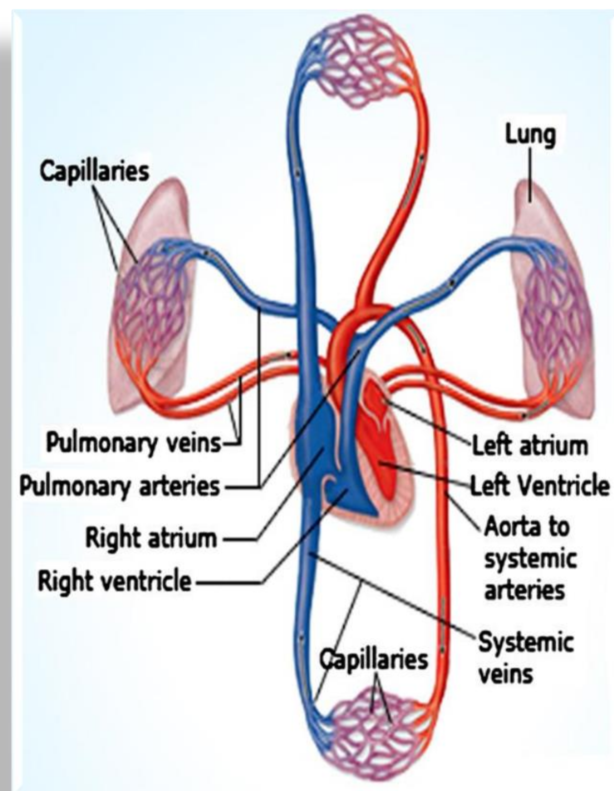
- **LEUKOCYTES-** are white blood cells. These are described as the *soldiers of the body* because their function is to defend the body against both infectious disease and foreign materials by *secreting antibodies.* Their life span can last for 3 days. The WBCs are colorless and have no nucleus.





- **THROMBOCYTES-** are the blood platelets. Compared to red blood cells, they are not true nucleus. Blood platelets are irregular in shape and their important attribute is they disintegrate upon contact with air. This is the reason why even in fresh blood samples, platelets cannot be observed. Their major function is to help in the clotting of blood and wounds and prevents excessive blood loss and the entry of bacteria.

CIRCULATION- the movement of blood through the vessels of the body induced by the pumping action of the heart.



TYPES OF CIRCULATION

1. **PULMONARY CIRCULATION.** Movement of blood from the heart, to the lungs, and back to the heart.
2. **CORONARY CIRCULATION.** Movement of blood through the tissues of the heart.
3. **SYSTEMIC CIRCULATION.** Movement of blood from the heart to the rest of the body, excluding the lungs

NAME: ____________________________________ YEAR AND SECTION :___________________

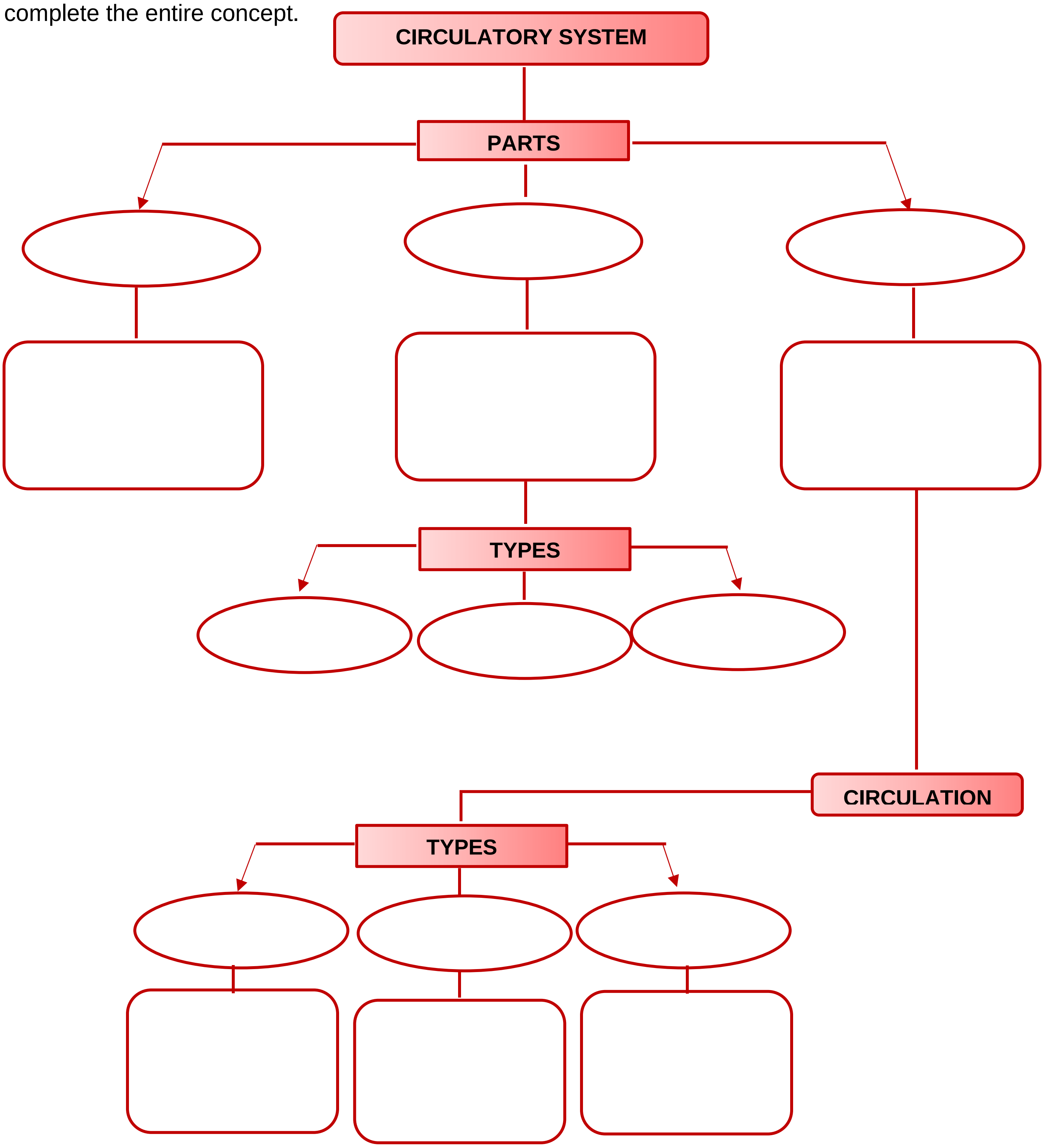
ACTIVITY 1: LET'S ORGANIZE!

OBJECTIVES:

- Identify the parts of the circulatory system.
- Explain the different types of circulation.

PROCEDURE:

Using the given graphic organizer, fill in the missing parts, description, and functions to complete the entire concept.



NAME: ____________________________________ **YEAR AND SECTION :**_______________

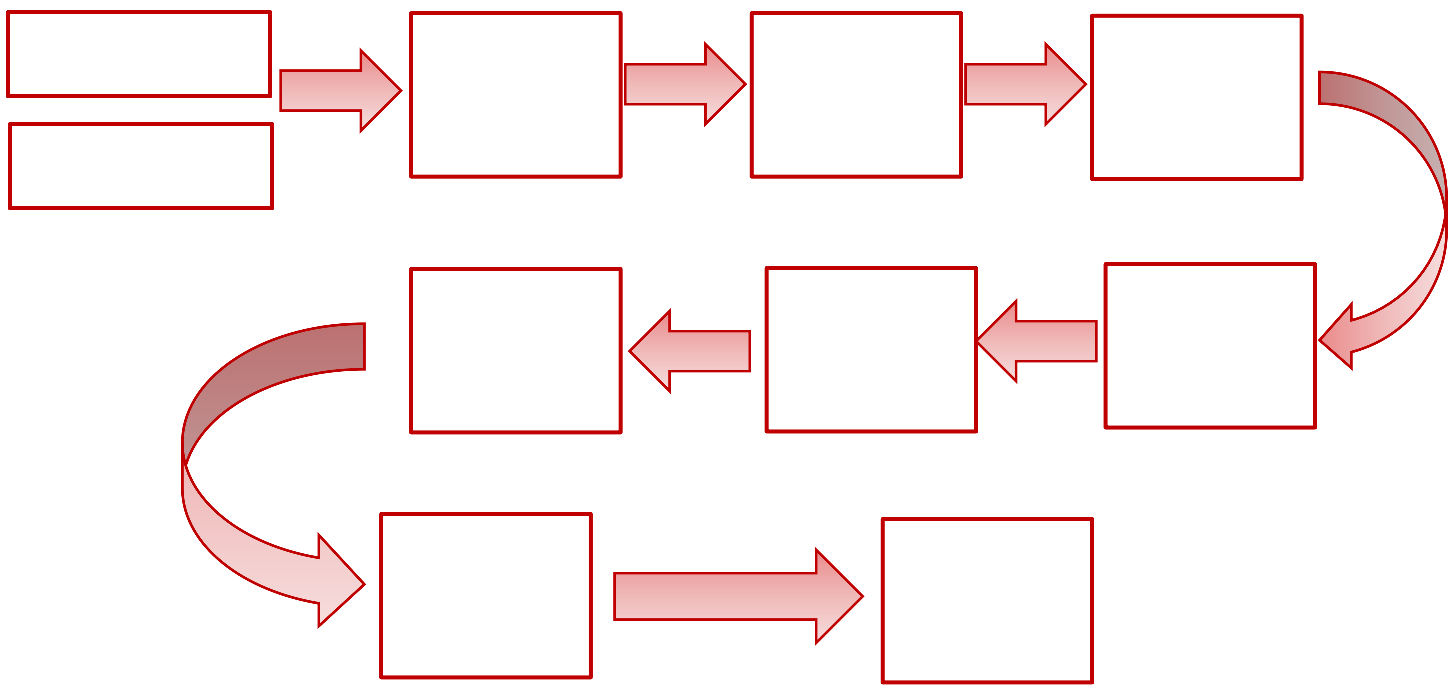
ACTIVITY 2: FLOW DIAGRAM

OBJECTIVE:

- Identify the flow of the blood within the blood vessels.

PROCEDURE:

- Trace the path of the blood from the heart to the lungs and to all parts of the body using the diagram below.



Concept Micro

The ***circulatory system*** is composed of the blood vessels, blood and the heart. The heart has different parts that help and control the flow of the blood and its distribution to different parts of the body. The blood vessels are arteries, capillaries and vein which give way to the passage of the blood. They are branched out all over the body. The blood constitutes plasma, platelets, red blood cells and white blood cells

TEST YOURSELF!

TEST I. <u>FILL IN THE BLANK</u> Supply the missing term to have functional understanding of the flow of blood in the pulmonary circulation. Get your answers from the key below. Some terms can be used twice in answering.

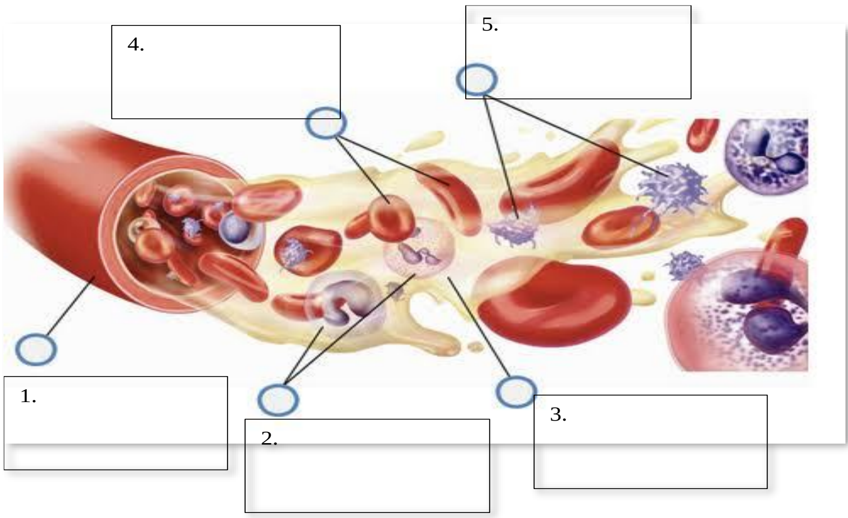
Tricuspid	right atrium	aorta	vena cava
All parts of the body	bicuspid	pulmonic	right ventricle
Aortic	left atrium	pulmonary veins	
Pulmonary artery	left ventricle	lungs	

Blood coming from all parts of the body is received by the 1.__________ of the heart by way of the blood vessel called 2.__________. Contraction of the right atrium sends blood to the 3.__________. At this point, the 4.__________valves are open. Contraction of the right ventricle forces blood to the 5.__________ for oxygenation by way of the 6.__________. At this point, the 7.__________ valves are open, while the 8.__________ valves are closed to prevent the back flow of blood in the right atrium.

From the lungs, oxygenated blood goes back to the 8.__________ of the heart via the blood vessels called 9.__________. Contraction of the left atrium forces blood to the 10.__________. At this point, the 11.__________ are open. Contraction of the left ventricle pumps blood to 12.__________ by way of the 13.__________, the largest artery. At this point, the 14.__________ valves are open while the 15.__________ are closed to prevent the backflow of blood in the left atrium.

TEST YOURSELF!

TEST II <u>IDENTIFICATION</u>. Identify the different components of the blood. Choose your answer on the box.



White Blood Cells
Plasma
Platelets
Blood Vessels
Red Blood Cells

TEST III. Complete the following sentence/s.

1. Red blood cells transport oxygen and ___________, ___________ to and from the cells. They also give blood its ___________ color.
2. Leukocytes ___________ any bacteria or viruses that enter the body. Vaccines help the body to make specific types of white blood cells to help us fight ___________.
3. Plasma a yellow liquid made up of 92% ___________. Plasma carries ___________ to the body and ___________ to the kidneys.
4. Platelets are particles that join together to stop ___________ repair and regenerate damaged tissue.
5. The three types of blood vessels are: ***Arteries*** that carry the blood with ___________ from the heart to the body, ***veins*** that carry the blood with carbon dioxide from the ___________ to the heart and ***capillaries*** with thin ___________ so gases and materials can pass through easily.

DEFINITION OF TERMS

GLOSSARY OF TERMS

AORTA- largest artery that conveys oxygen- rich blood pumped by the left ventricle to all parts of the body.

ARTERIOLES- The arterioles are smallest arteries.

ARTERY- convey oxygen- rich blood away from the heart.

BLOOD –is a liquid tissue and the circulating medium of the cardiovascular system.

BLOOD VESSELS- convey blood to different parts of the body.

CAPILLARIES- are the smallest and most numerous blood vessels

CIRCULATION- the movement of blood through the vessels of the body induced by the pumping action of the heart.

HEART – the pumping organ of the cardiovascular system.

PLASMA- is the liquid portion of the blood.

VEINS - carry deoxygenated blood to the heart.

REFERENCES

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ELECTRONIC SOURCES

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